Finite-Memory Projection Corrections In Cosmological Consistency Diagnostics

Abstract

We introduce a bounded finite-memory projection operator for diagnostic cosmological consistency tests. The operator is formulated as an effective projection response over normalized depth, `W(x)=1+rho*x^3`, with a passive memory-budget bound `rho<=4`. The construction is not presented as a complete cosmological model or an observational detection. It is instead a disciplined diagnostic module for comparing projected response shapes against backreaction-aware distance consistency reconstructions. In a BAO-only diagnostic reconstruction, the locked K2 window overlaps the full target envelope. In an SN+BAO source-split stress test, strict median-sign matching exhibits a localized tension; a reconstruction-aware sign-stability gate resolves this tension without widening the K2 window. The result motivates a future full-likelihood comparison while preserving explicit claim boundaries.

1. Introduction

Modern cosmological consistency tests increasingly distinguish background expansion, light-propagation effects, curvature consistency, and cosmic backreaction. A residual relative to a homogeneous FLRW reference model is therefore not automatically evidence for new physics: it may arise from optical sampling, reconstruction convention, averaging, or the choice of observable combination.

Heinesen and Clifton recently emphasized that broad classes of cosmological models can be separated by how they violate FLRW curvature-consistency tests. They isolate two especially relevant routes: deviations in optical properties and deviations in large-scale expansion. Koksbang then applied this kind of observable logic to constrain cosmic backreaction over an extended redshift range, finding consistency with vanishing backreaction at one standard deviation while also emphasizing that significant backreaction is not yet ruled out. These papers are useful here because they make the main methodological lesson sharp: a projected cosmological residual must be tested against optical, geometrical, and reconstruction degeneracies before it is interpreted.

This note develops a deliberately narrow finite-memory projection diagnostic that can be placed alongside such tests. The objective is not to disclose or validate a full underlying theory. Instead, we ask whether a predeclared, bounded projection-memory operator can define a reproducible diagnostic window and survive basic source-split stress tests.

2. Minimal Effective Disclosure

Only the effective module needed for this note is used. We assume that an observed consistency residual can be represented as a projected response over a normalized depth coordinate,

```
0 <= x <= 1.
```

The variable `x` should be read operationally: it orders the accumulated projected response along the diagnostic light-path/reconstruction depth. It is not a claim that the full underlying theory has been reconstructed from the paper.

The finite-memory correction is a multiplicative response operator acting on a baseline projected shape:

```
K2(x) = W(x) K1(x).
```

This note studies the locked K2 form

```
W(x) = 1 + rho*x^3.
```

The form is intentionally simple. Its purpose is to test whether a bounded, delayed, monotone memory response can be made reproducible before comparison with diagnostic envelopes.

3. Passive Memory Bound

The passive memory-budget condition requires the accumulated tail response not to exceed the unit local response budget. For the cubic operator,

```
E_tail = Integral_0^1 rho*x^3 dx = rho/4.
```

The unit baseline budget is

```
E_0 = Integral_0^1 1 dx = 1.
```

Passivity requires

```
E_tail <= E_0,
rho/4 <= 1,
rho <= 4.
```

This is the key discipline of the construction: the memory depth is bounded before the diagnostic comparison. The bound is not chosen by looking for the best residual fit.

4. Kernel-Shape Selection

The cubic kernel is selected from the power-kernel family

```
W_p(x) = 1 + rho*x^p.
```

For this family, the passive memory rule generalizes to

```
Integral_0^1 rho*x^p dx <= Integral_0^1 1 dx,
rho/(p+1) <= 1,
rho <= p+1.
```

Two additional effective-operator rules select the usable shape:

```
low-depth invisibility: Delta W_p(0.25) = (p+1)*0.25^p <= 0.10
no endpoint spike:     Integral_0.75^1 (p+1)*x^p dx <= 0.75
```

The first rule rejects kernels that disturb the already locked low-depth response too early. The second rejects kernels whose memory budget collapses too sharply onto the last quarter of the path. In the present audit, p=1 and p=2 are too visible at low depth, while p>=4 is too endpoint-concentrated. The cubic p=3 case is the remaining admissible power-kernel representative.

The locked diagnostic window is therefore

```
W(x) = 1 + rho*x^3,
eta in [0.25, 0.50],
rho in [3.00, 4.00].
```

5. Diagnostic Targets And Gate Policy

The targets used here are reconstructed diagnostic envelopes rather than direct published likelihoods. The comparison is therefore deliberately separated into three levels:

```
sign-stable reconstructed point   -> exact sign gate
sign-unstable reconstructed point -> envelope / diagonal residual gate
controls and extrapolations       -> stress diagnostics only
```

A reconstructed point is sign-stable only when the relevant method/degree families agree on the sign of the diagnostic median. If the reconstruction sign is method-sensitive, the median sign is not used as a hard falsifier. Instead, the locked K2 prediction must remain inside the diagnostic envelope and have a normalized residual not exceeding one standard diagnostic width.

This policy is not a substitute for a full likelihood. It is a triage rule that prevents a method-sensitive reconstructed median from being overinterpreted.

6. Results

The minimal result summary is:

Diagnostic	Result	Interpretation
BAO-only window overlap	1.0000000000	Locked K2 overlaps the BAO-only diagnostic envelope at all tested points.
BAO-only best curve overlap	1.0000000000	The best locked K2 BAO-only curve remains inside the envelope.
SN+BAO sign-stable pass fraction	1.0000000000	All sign-stable SN+BAO points pass exact sign matching.
SN+BAO sign-unstable likelihood pass fraction	1.0000000000	All sign-unstable SN+BAO points pass envelope / diagonal residual handling.

The strict SN+BAO median-sign test exhibits a localized tension near `z=1.1925`. The diagnostic audit shows that this point is method-degree sensitive: lower-degree reconstructions are positive, while higher-degree reconstructions pull the median negative. The locked K2 prediction does not follow the negative median sign at that point, but it remains inside the broad diagnostic envelope with a small normalized residual. Under the sign-stability gate, this is treated as a likelihood/envelope pass rather than a hard sign-failure.

The current status is therefore:

```
BAO-only: diagnostic pass
SN+BAO strict median-sign: localized warning
SN+BAO sign-stability-aware gate: diagnostic pass
claim level: theory-method diagnostic, not observational discovery
```

7. Expected Significance

The main significance of the construction is not that it proves a new cosmological model. Its value is methodological:

- it gives a compact finite-memory operator whose amplitude is bounded before

comparison with data;

- it interfaces naturally with FLRW consistency, optical-propagation, and

backreaction tests;

- it demonstrates a way to handle sign-unstable reconstructed diagnostics

without silently overclaiming;

- it provides a small public module that can be tested independently of the

full underlying theory program.

The near-term audience is likely narrow: researchers interested in cosmological consistency tests, backreaction, light propagation, and method-level modified-gravity phenomenology. The higher-impact route requires a later full covariance or shrinkage-covariance likelihood comparison with direct public likelihood products.

8. Limitations

This manuscript is a theory-method diagnostic note. It does not claim:

- a full covariance likelihood;
- a direct observational detection;
- exclusion of cosmic backreaction;
- a complete derivation of cosmology from an underlying action;
- disclosure or reconstruction of a full Tau Core theory.

The next stage is a paper-aligned likelihood comparison using covariance or shrinkage-covariance products and direct published likelihood data where available.

References

- A. Heinesen and T. Clifton, "Observational Tests for Distinguishing Classes of Cosmological Models", arXiv:2604.07244, 2026, https://arxiv.org/abs/2604.07244.

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- C. C. Dyer and R. C. Roeder, "The Distance-Redshift Relation for Universes with no Intergalactic Medium", Astrophysical Journal, 1972.